Supplementary Methods and Results

Graph-Based Evidence Recovery for Surgical Question Answering

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Study Materials

The recovery study includes 15 published sources and one question per source (Table S1). The same pairs were used at final evidence limits of 800, 1,600, and 3,200 characters. The smaller-limit analysis is exploratory and does not constitute a new independent test set. Four development pairs were excluded. A related ERAS Emergency Laparotomy article, PMC10241558, was excluded from the registered recovery panel without replacement before method outputs.

The source-based checklist covers the first ten articles in Table S1. These articles contain the 26 propositions and 36 attached qualifications assessed in that checklist. The questions below are transcribed from the retained recovery records. Source excerpts are limited to 14,000 characters and divided into 400-character chunks; full source text and the excerpt used for generation are distinct inputs.

Table S1: Source articles and recovery-study questions. Article links use PubMed Central identifiers.

Source	Question
PMC9359950	How did the presence of ctDNA after surgery relate to prognosis and prediction of adjuvant immunotherapy benefit in the Impower010 study?
PMC8987560	Which three standardized indices or scales does the source describe for assessing surgical invasiveness or bleeding severity in spine surgery, and in what setting do they tend to be used?
PMC8088961	In perioperative management of severe irAEs, what lack-of-response interval to glucocorticoid treatment permits starting a TNF-α inhibitor immediately?
PMC11606666	Which three subthemes organize the source's wound-healing recommendations after thoracic aortic surgery, and how do the aims differ between recommendations after sternotomy and lateral thoracotomy?
PMC11434558	According to the source, when should air-cleaning technologies such as HEPA filtration be implemented in facilities lacking airborne infection isolation rooms?
PMC9297612	What were the two volume allowances for the carbohydrate-enriched beverage on the night before and the day of surgery, and what timing restriction applied to the day-of-surgery allowance?
PMC11213574	When should vocal fold mobility be assessed before contralateral thyroid ablation following ipsilateral ablation?
PMC9879827	Which three methods are distinguished for quantitative tissue perfusion assessment using ICG NIR fluorescence imaging, and what reference is used to normalize fluorescence intensity in the normalized method?

Source	Question (continued)
PMC12094288	Why are the ATP-3, Framingham and CHADS-VASc scores unsuitable for predicting cardiovascular risk during the perioperative period of non-cardiac surgery? (English translation of the archived question; the original experimental input is unchanged.)
PMC11634417	Which three methods does the source list for controlling larger vessels, bile ducts, and the hepatic pedicle during robotic hepatectomy, and what factors determine the choice?
PMC11393364	In surgically remediable epilepsy cases, what does the source state about cenobamate use and imminent surgery, and under what condition may a patient elect to defer or cancel surgery?
PMC8725119	Which two instrumentation approaches did experts prefer for segmentation and delamination in diabetic vitrectomies, and what qualification did the source report about consensus on their use?
PMC9579840	According to the source, why should the cited meta-analyses not be used to justify downgrading the class of recommendation for CABG in patients with stable ischemic heart disease?
PMC9287504	What three main techniques can be used to create the femoral tunnel in ACL revision surgery, and what qualification does the source give about their comparative superiority?
PMC8397635	In the consensus, when was wound re-evaluation supported after starting antibiotic treatment for surgical site infections, and under what condition was changing antibiotic treatment supported?

Additional Answer Outcomes

The initial 3,200-character comparison used five methods with each generator. All larger-model methods produced 15 citation-supported complete answers. No smaller-model method produced a valid delivered answer under the required answer-or-abstain format (Table S2). These results should not be read as absence of all supported content: each smaller-model method produced one nonempty answer text judged source-complete, but its decision field was invalid. Invalid decisions were not repaired or resampled.

The reduced-budget full-source outcomes, delivery counts, and citation-supported complete answer counts are reported in main Table 2. There are 15 questions per method and limit; multiple method records do not increase the number of independent questions. Identical final evidence could reuse the same generated response. Text planning at smaller limits reused the full source index, extracted propositions, and initial answer from the recovery study. Consequently, these limits concern final generation context and do not bound all upstream information.

Source-Based Fidelity Assessment

The automatic assessment of the recovery panel covered 40 propositions and 53 qualifications, all labeled faithful, with no requested information marked missing for any of the 15 sources. This assessment is distinct from the completed checklist supplied by Manglam Srivastav.

Table S2: Initial 3,200-character outcomes. Every row uses 15 questions. The first column distinguishes the generator, not the extraction or planning model.

Generator	Method	Delivered	Supported complete	Abstained	Invalid decision
Larger	Lexical	15	15	0	0
Larger	Generic recovery	15	15	0	0
Larger	Text planning	15	15	0	0
Larger	Graph recovery	15	15	0	0
Larger	Qualification ablation	15	15	0	0
Smaller	Lexical	0	0	14	1
Smaller	Generic recovery	0	0	12	3
Smaller	Text planning	0	0	13	2
Smaller	Graph recovery	0	0	14	1
Smaller	Qualification ablation	0	0	14	1

The latter checklist contains 26 proposition judgments, 36 qualification-attachment judgments, and ten question-level coverage checks. Each qualification has a separate judgment, explanation, and supporting quotation, linked to a parent proposition. All 62 proposition/qualification judgments are labeled faithful; all ten coverage judgments record no missing required information. Expert knowledge beyond the source is marked absent throughout. These labels describe source correspondence, not clinical correctness.

The original template contains the 26 proposition rows and ten coverage rows. Those rows match the expanded 72-item checklist. The remaining 36 identifiers correspond to the attachment map, whose judgment and source-quotation fields agree with the expanded file. Locally assigned attachment identifiers support traceability and are not recovered original graph-edge identifiers. The three CSV files accompanying this supplement preserve those records without alteration.

The Portuguese cardiovascular source was assessed with automated assistance, explicitly identified in its checklist entry. The records do not document separate human checking for every judgment and contain no review-date or AI-model/version fields. No independent human, bilingual, or surgeon validation is inferred. Explicit dependency edges were not assessed, and the checklist covers only ten of the 15 sources. The original surgeon ratings and original manual-curation history are not established by this assessment.

Earlier Verification Experiments

Table S3 retains the principal null and adverse findings from the earlier candidate-verification experiments. Those experiments used different sources, policies, and outcome definitions from evidence recovery. Their results are not pooled with the 15-question study, and changes in verdict accuracy are not treated as improvements in generated-answer quality. The reported intervals and tests below are retained historical analyses; they were not recalculated from aggregate counts for the present descriptive comparison.

Table S3: Earlier findings relevant to interpretation of the graph contribution. "pp" denotes percentage points.

Comparison	Reported finding
Candidate verification: graph versus strengthened text	Accuracy difference -1.67 pp; 95% source-cluster interval -4.44 to 1.11; exact two-sided $p = 0.4531$.
Natural-answer assessment	Useful complete-answer yield was identical; graph and strengthened text each accepted 17 known errors.
Separate atomic verification, larger model	Graph and text accuracy were identical.
Separate atomic verification, smaller model	Graph corrected five cases relative to text ($p = 0.0625$): three source-support classification errors and two unresolved judgments without valid supporting quotations. Natural-answer yield was unchanged.
Equivalent tabular execution	The same selection operations reproduced the graph-selected source passages. This does not support an effect attributable to graph storage alone.
Earlier source audit	Correct quotation provenance, coverage of required information, and faithful representation of relationships were distinct; valid quotations did not guarantee adequate task coverage or correctly represented relations.

Statistical Reporting

Main Table 2 and Figure 3 use per-arm Wilson score intervals based on the reported strict answer count k and $n = 15$. With $\hat{p} = k/n$ and $z = 1.959963984540054$, the interval is

$$\frac{\hat{p} + z^2/(2n) \;\pm\; z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}.$$

For example, 13/15 gives 86.7% (95% interval 62.1–96.3), while 15/15 gives 100.0% (79.6–100.0). These intervals treat a question as one observation and do not adjust for related-source dependence. They are not paired confidence intervals for differences between methods. The marginal reduced-budget counts do not identify discordant question pairs, so paired significance tests are not reconstructed from them. Overlapping or identical intervals do not establish equivalence.

Reference-span coverage and requirement coverage are process measures. Their aggregate percentages are reported without reconstructed source-level error bars. In Figure 2, text-planning context whiskers use the reported interquartile range of 400–800 characters; they are not confidence intervals. Final evidence length excludes the full-source extraction and planning stages and cannot establish total latency, monetary cost, or operational efficiency.